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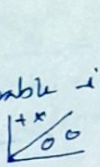
Support Vector Machine

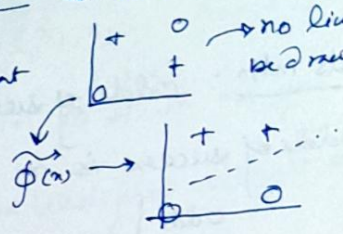
1) Our Assumption was Data was linearly seperable i.e. it is possible to draw a line between two class.



a) optimization maximum margin b) decision rule

2) When data is not linearly seperable.

We try to apply here $\phi(x)$ (Transform) so that the data will be change. so make the data seperable.



This transformations are like kernel function

- Project the data to a space that is seperable
- Transform the points to a space s.t. they are seperable.
- Let the transform f^n be $\phi(x_i)$ on a sample
- Both the decision rule and optimization depends on dot product of samples.
- So we need to optimize the dot product $\phi(x_i)\phi(x_j)$
  ↳ To do this we use kernel function $K(x_i, x_j) = \phi(x_i)\phi(x_j)$
- The kernel function provides the dot product
- Typical kernel functions are

1) $(\vec{u}\cdot\vec{v}+1)^n$. 2) $\frac{-\|\vec{x_i}-\vec{x_j}\|}{\delta}$ etc.

Supervised Classification : Logistic Regression

$(y_i, \vec{x_i})$: x_i independent, y_i is categorical.

Binary classification Problem where number of classes is 2, i.e. either 0 or 1 (y)

- Use probabilities to classify using a threshold, given a new data point.

Model:

- make probability $p(x)$ a linear function of x

$p(x) = \beta_0 + \beta_1 x$ → it can go beyond 0 to 1

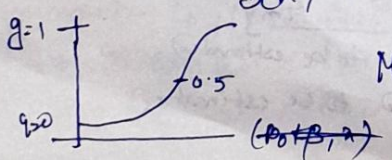
- To make $p(x)$ unbounded below 0 & above 1.
- Make $\log(p(x))$ a linear function of x.

$\ln(p(x)) = \beta_0 + \beta_1 x$ ⇒ range 0 to +ve ∞

- but for probability we need to bound between 0 to 1.

Thus we use Sigmoid Function S shaped curves which maps x to a value between 0 & 1.

$$S(x) = \frac{1}{1+e^{-x}}$$



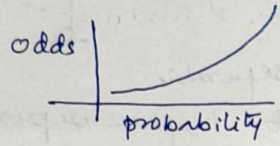
$$\text{Model } p(x) = \frac{e^{(\beta_0+\beta_1 x)}}{1+e^{(\beta_0+\beta_1 x)}}$$

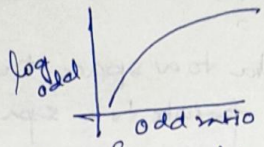
if $\beta_0 + \beta_1 x = -\infty$ $= \frac{e^{-\infty}}{1+e^{-\infty}} \Rightarrow 0$

if $\beta_0 + \beta_1 x = +\infty$, $\frac{e^{\infty}}{1+e^{\infty}} = 1$

<u>Odds ratio</u>: odds of success $= \frac{\text{probability of success}}{\text{probability of failure}}$

The odds of success is r to 1.





- <u>log of odds ratio</u>: greater the odd ratio logs odds is also greater.

→ maps probability (0 to 1) to log odds from $-\infty$ to $+\infty$ — <u>logit transformation</u>.

→ <u>Model</u> $\log\left(\frac{P(x)}{1-P(x)}\right) = \beta_0 + \beta_1 x$ —— (eq1)

- logistic regression models probility of default class i.e. model the probability that an input x belong to class $y=1$.

$$p(x) = P(y=1 \mid x)$$

How we obtain $\log\left(\frac{P(x)}{1+P(x)}\right) = \beta_0 + \beta_1 x$

- Remember that, $P(x) = \frac{e^{(\beta_0+\beta_1 x)}}{1+e^{(\beta_0+\beta_1 x)}}$ {using Sigmoid fun}

$$1 - P(x) = 1 - \frac{e^{(\beta_0+\beta_1 x)}}{1+e^{(\beta_0+\beta_1 x)}} = \frac{1+\cancel{e^{(\beta_0+\beta_1 x)}} - \cancel{e^{(\beta_0+\beta_1 x)}}}{1+e^{\beta_0+\beta_1 x}}$$

$$= \frac{1}{1+e^{\beta_0+\beta_1 x}}$$

<u>Substituting</u>,

$$\log\left(\frac{P(x)}{1-P(x)}\right) = \log_e\left[e^{\beta_0+\beta_1 x}\right]$$

$$= \beta_0 + \beta_1 x$$

<u>Estimate the parameters $\hat{\beta}_0$ & $\hat{\beta}_1$</u>

Find $\hat{\beta}_0, \hat{\beta}_1$ such that model equation gives the best possible classification for the input from both classes

⇒ Use maximum likelyhood estimation. (MLE)

<u>MLE</u> (say): Let $x_1, x_2, x_3 \ldots x_n$ have joint probability mass function $f(x_1, x_2, \ldots x_n; \underbrace{\theta_1, \theta_2, \ldots \theta_m}_{\text{Parameter to be estimated}})$.

Where $\theta_1, \theta_2, \theta_3 \ldots \theta_n$ are unknown & need to be estimated

when $x_1, x_2, \ldots x_n$ are observed sample values. The function of is called the likelihood function

→ The maximum likelihood estimates $\hat{\theta}_1, \hat{\theta}_2 \ldots \hat{\theta}_m$ are those values of θ_i that maximizes the likelihood function so that

$$f(x_1, x_2 \ldots x_n, \hat{\theta}_1, \ldots \theta_2, \ldots \hat{\theta}_m) \geq f(x_1, x_2 \ldots x_m, \theta_1, \ldots \theta_m) \quad \forall \theta_i$$

Example: Suppose we are ~~having~~ tossing a coin 10 times we got the output H H T H H H T H T H

#T : 3 #H : 7 Maximum Likelihood estimate of the parameter p : probability of getting heads.

$$f(x_1, x_2, \ldots, x_{10}; p) = \underbrace{p \cdot p \cdot (1-p)\, p\, p\, p\, (1-p)\, p\, (1-p)\, p}_{10}$$

$$= p^7 (1-p)^3 \quad \text{—— (3)}$$

~~max~~ maximize f_i

to simplify use log

take log on equation (3)

$$\ln f(\ ) = 7 \ln p + 3 \ln (1-p)$$

$$\frac{\partial f}{\partial p} = 7/p - \frac{3}{1-p} \quad \text{— (4)}$$

equation (4) to zero $\frac{7}{p} - \frac{3}{1-p} = 0 \Rightarrow p = \frac{7}{10} = 0.7$

⇒ How likely the observed sample is ~~rea~~ function of possible parameter value

i.e. the parameter value that agrees most closely with the observed data: → (maximizing the likelihood)

Logistic Regression Model

Using MLE, estimate $\hat{\beta}_0, \hat{\beta}_1$ that would result in a model that predict a value very close to 1 for default class & very close to 0 for other class i.e MLE solves the condition $\sum_i y - P(y=1) \mid x_i = 0$

Likelihood function $L(x_i, \beta_0, \beta_1)$

$$= \prod \cancel{\text{probability}}\ p(x_i)^{y_i} (1-p(x_i))^{(1-y_i)}$$

$$\left[y_i = \begin{cases} 0 \\ 1 \end{cases} \text{ the two classes} \right]$$

Take log on $L(x_i, \beta_0, \beta_1)$

$$\Rightarrow \log(L(x_i, \beta_0, \beta_1)) = \log\left(\prod (p(x_i))^{y_i} (1-p(x_i))^{(1-y_i)}\right)$$

$$= \sum y_i \log p(x_i) + (1-y_i) \log(1-p(x_i))$$

maximize $\log(L(x_i, \beta_0, \beta_1))$ [Now it's a non linear function]

→ we can use several non linear optimization method.

→ one such method gradient descent Method

→ obtain β_0 & β_1 using this method

Once $\hat{\beta}_0, \hat{\beta}_1$ are estimated, use the logistic model

$$P(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$ to predict the class given an

input x. If $\beta_0 + \beta_1 x$ is non-negative then we get

a value $P(x) > 0.5$ so predict $y = 1$ (default class)

else $P(x) < 0.5$ predict $y = 0$ (the other class)

— <u>Decision Boundary</u> is the equation $\beta_0 + \beta_1 x$

if $\beta_0 + \beta_1 x = 0 \Rightarrow P(x) = \frac{e^0}{1 + e^0} = \frac{1}{2} = 0.5$